



QS REIMAGINE EDUCATION 2026 | SUPPORTING EVIDENCE

Scalability and Financial Sustainability

A milestone-led plan for production, institutional adoption and responsible growth



1. Basis of this evidence

This paper is a planning document, not a set of audited accounts. It distinguishes money already secured from funding still to be raised, and it labels forecasts as management assumptions. Revenue has not been presented as historical income.

CASH SECURED

\$6,740

\$5,240 university support plus a \$1,500 EdUHK Edventures cash award.

NON-CASH SUPPORT

\$5,500

AWS credits awarded through EdUHK Edventures. These are not treated as cash.

REPORTED FIELD REACH

33+ schools

Prototype and pilot activity reported across six Indian states.

How to read the numbers

Component costs are current planning estimates and must be replaced by supplier quotations before a production commitment. Sales, contracts and operating results are a base case forecast. Award letters, university records, pilot logs, quotations and purchase orders should be retained as the underlying evidence file.



2. Why the product can scale

TACTO keeps the expensive electronics in one reader and keeps each coding block passive. A block contains a tactile form, a Braille label and an NFC tag. It has no battery, processor or radio. New commands and lessons can therefore be added through a new block shell, NFC mapping, audio file and curriculum activity without redesigning the complete kit.



Modular hardware

A lost or damaged block can be replaced individually. The reader remains in service, which reduces maintenance cost and downtime.



One platform, several levels

The same kit can progress from sequencing to loops, conditions, variables and functions through software and curriculum updates.



Low-cost localisation

Language changes affect audio, labels and lesson material. They do not require a new electronic reader design.



Distributed delivery

Standard parts, documented assembly and replaceable components allow production to move from university prototyping to contract manufacturing or approved regional assembly.

3. Unit economics

The reference classroom kit contains one reader and 20 tactile coding blocks. The pilot estimate uses current retail component prices plus explicit allowances for assembly, quality checks, scrap, replacements and freight.

All figures are in US dollars and rounded for planning.

Component group	Quantity	Estimate
Raspberry Pi Zero 2 W	1	\$22
PN532 NFC reader modules	2	\$6
NTAG213 NFC tags	20	\$3
Audio amplifier and speaker	1 set	\$2
Battery and charging module	1 set	\$2
32 GB microSD card	1	\$9
Printed enclosure and blocks	1 set	\$13
PCB, wiring and connectors	Allowance	\$4
Packaging and printed guide	Allowance	\$3
Raw component estimate		\$64
Assembly, quality, scrap, replacements and freight	Allowance	\$12
Pilot-stage landed cost		\$76

Pricing and margin discipline

The base case uses a \$149 list price and lower average realised prices for institutional and volume orders. A sale should not be accepted below the approved margin floor unless a grant or CSR contract explicitly pays the difference.

Stage	Landed cost	Average price	Gross margin	Basis
Pilot batch	\$76	\$139	45%	Retail parts and manual assembly
Year 2 batch	\$60	\$135	56%	Supplier pricing and repeatable assembly
Year 3 batch	\$52	\$129	60%	Contract manufacturing and volume buying



4. Commercial model and base case forecast

Kit sales alone are not expected to fund educator onboarding, curriculum adaptation or field support. The model therefore uses four linked income streams: hardware, paid implementation and educator training, sponsored deployment contracts, and recurring support or content licensing.

Hardware

\$149 list price, with average realised prices of \$139, \$135 and \$129 in Years 1 to 3.

Implementation

A separate institutional fee for readiness review, educator training, setup and initial delivery support.

Sponsored deployments

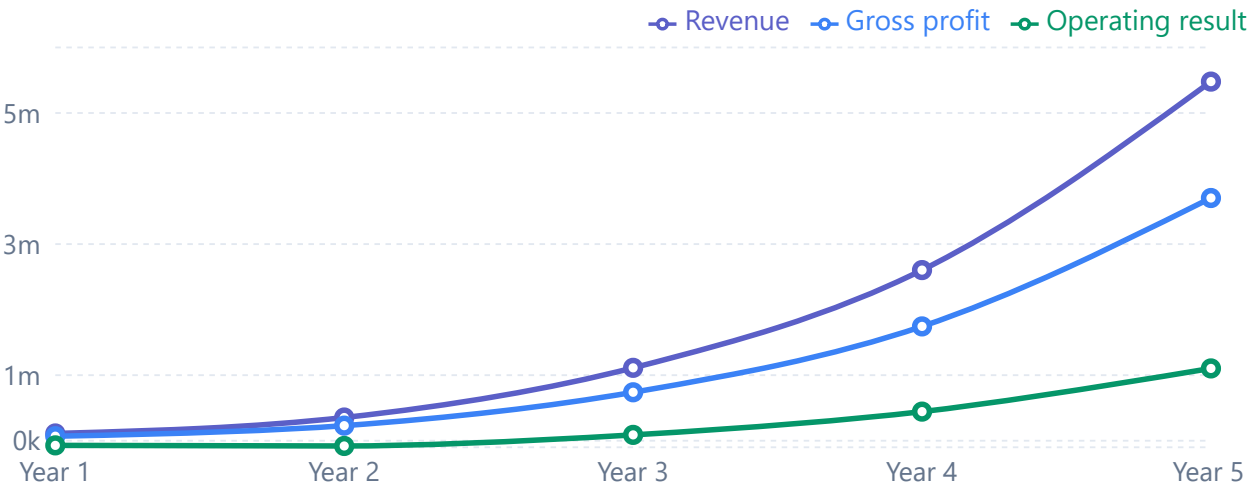
Grant, CSR and research contracts that purchase defined deployments for schools unable to pay directly.

Recurring income

Annual support, replacement coverage, dashboard services, curriculum packs and localisation licensing.

Five-year base case

Revenue and results in US dollar thousands. Operating result is before tax, interest and depreciation.



Assumption	Year 1	Year 2	Year 3
Kits sold	250	900	3,000
Kit revenue	\$34.8k	\$121.5k	\$387k
Implementation and training	\$36k	\$112.5k	\$300k
Sponsored deployment contracts	\$40k	\$96k	\$225k
Support, content and licensing	\$0k	\$24k	\$200k
Gross profit	\$69k	\$230k	\$739k
Operating result	\$-71k	\$-80k	\$89k
Total revenue	\$110.8k	\$354k	\$1.1m

Break-even logic

The base case reaches a positive operating result in Year 3, after annual volume reaches about 3,000 kits and service income covers the delivery team. Hiring and inventory are staged against contracted demand, so growth can be slowed if conversion takes longer.

Downside response

If unit sales are 30% below plan, TACTO will defer permanent hires, use smaller production batches, prioritise paid institutional deployments and keep new country entry on hold. The product architecture does not require a large factory commitment.



5. Funding plan and sustainability

Existing deployable cash supports the controlled pilot batch, but it does not fund tooling, inventory, compliance work and the delivery team required for commercial scale. The next target is a \$200,000 pre-seed round. Release of spending is tied to technical and commercial milestones.

TARGET RAISE

\$200,000

Indicative planning target. No future funding is shown as secured.

Use	Amount	Share
Product engineering, tooling and safety validation	\$75,000	37.5%
Inventory and working capital	\$50,000	25.0%
Structured pilots and implementation	\$35,000	17.5%
Curriculum, accessibility review and localisation	\$25,000	12.5%
Company setup, IP and contingency	\$15,000	7.5%

Diversification plan

Earned income

Hardware margin, implementation fees, educator training, support and curriculum services become the main source of operating sustainability.

Non-dilutive support

Education grants, research collaborations and competition awards fund validation, evidence generation and open educational resources.

CSR and government procurement

Funders purchase defined school deployments with agreed delivery, training and outcome reporting, rather than providing unrestricted subsidy.

Impact investment

Equity or a convertible instrument funds tooling, working capital and team capacity after the reference design and institutional demand are validated.

6. Growth plan and decision gates

Scale is treated as a sequence of evidence gates. Production, hiring and geography expand only after the preceding stage is shown to work.

Lock the reference design

0 to 6 months

Confirm the bill of materials, obtain supplier quotations, complete durability and safety checks, and freeze the Basic and Explorer kit specifications.

- ✓ **Decision gate:** No production increase until scan reliability, audio response, setup time and replacement rates meet the pilot thresholds.

Run a controlled production batch

6 to 12 months

Produce up to 35 kits for structured deployments, document teacher onboarding, and collect comparable learner and product data at each site.

- ✓ **Decision gate:** Proceed when schools can run sessions using the guide and training package without continuous founder support.

Convert pilots into paid institutional use

12 to 24 months

Move to 250 to 900 annual kit sales, introduce paid implementation packages, establish repair and replacement procedures, and secure a manufacturing partner.

- ✓ **Decision gate:** Add inventory only against purchase orders, sponsored deployment contracts, or a defined distributor forecast.

Build regional delivery capacity

Years 3 to 5

Use contract manufacturing, educator certification, regional distributors and local content partners to serve India first, followed by selected Asian and UK markets.

- ✔ **Decision gate:** Enter each new market only after local curriculum mapping, accessibility review, after-sales ownership and channel economics are confirmed.

Conclusion

TACTO can scale because the product is modular, the electronics are concentrated in one reader, and content can be localised without rebuilding the hardware. Financial sustainability does not depend on selling low-margin kits alone. The plan combines hardware income with implementation, sponsored deployments and recurring support, while using grants and investment for the stages they are best suited to fund. The forecast is ambitious, but every increase in production is tied to a clear technical or commercial gate.

Source note: Project TACTO management planning model, 11 August 2026. Current support figures are based on project records. Pilot reach is reported by the project team. Forecasts are planning assumptions and should be refreshed when supplier quotations, signed orders and audited accounts become available.